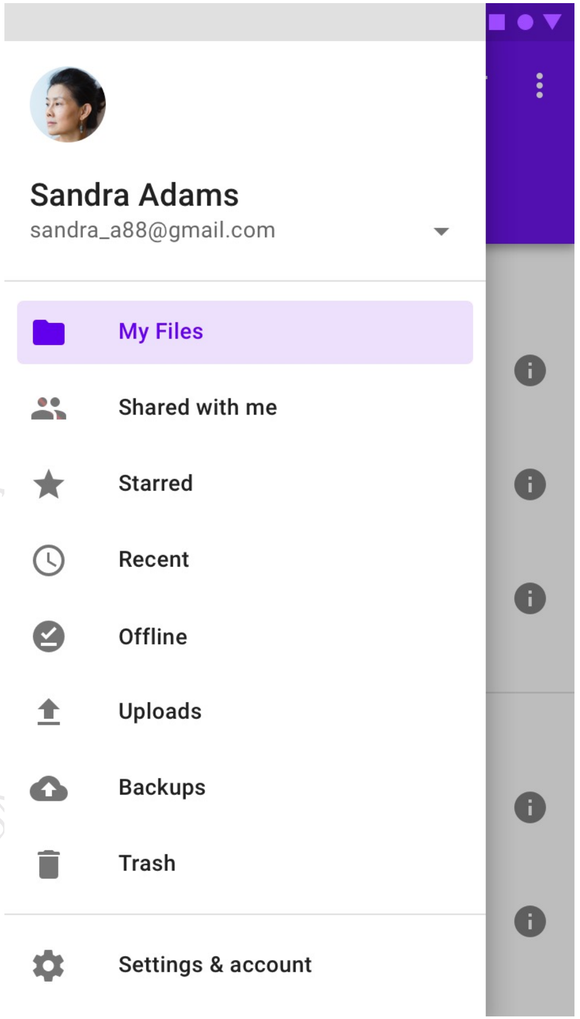


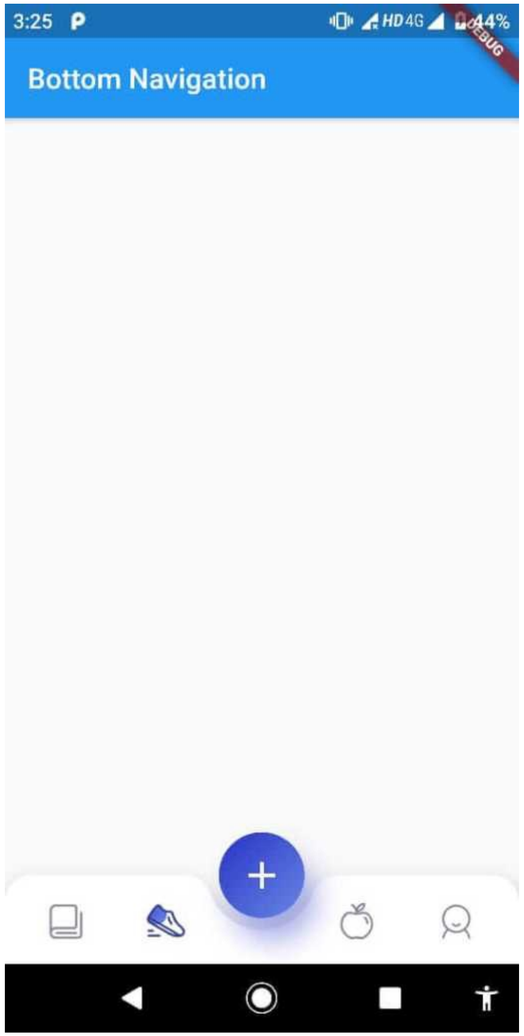
Навигация в приложении

МИРЭА

Введение

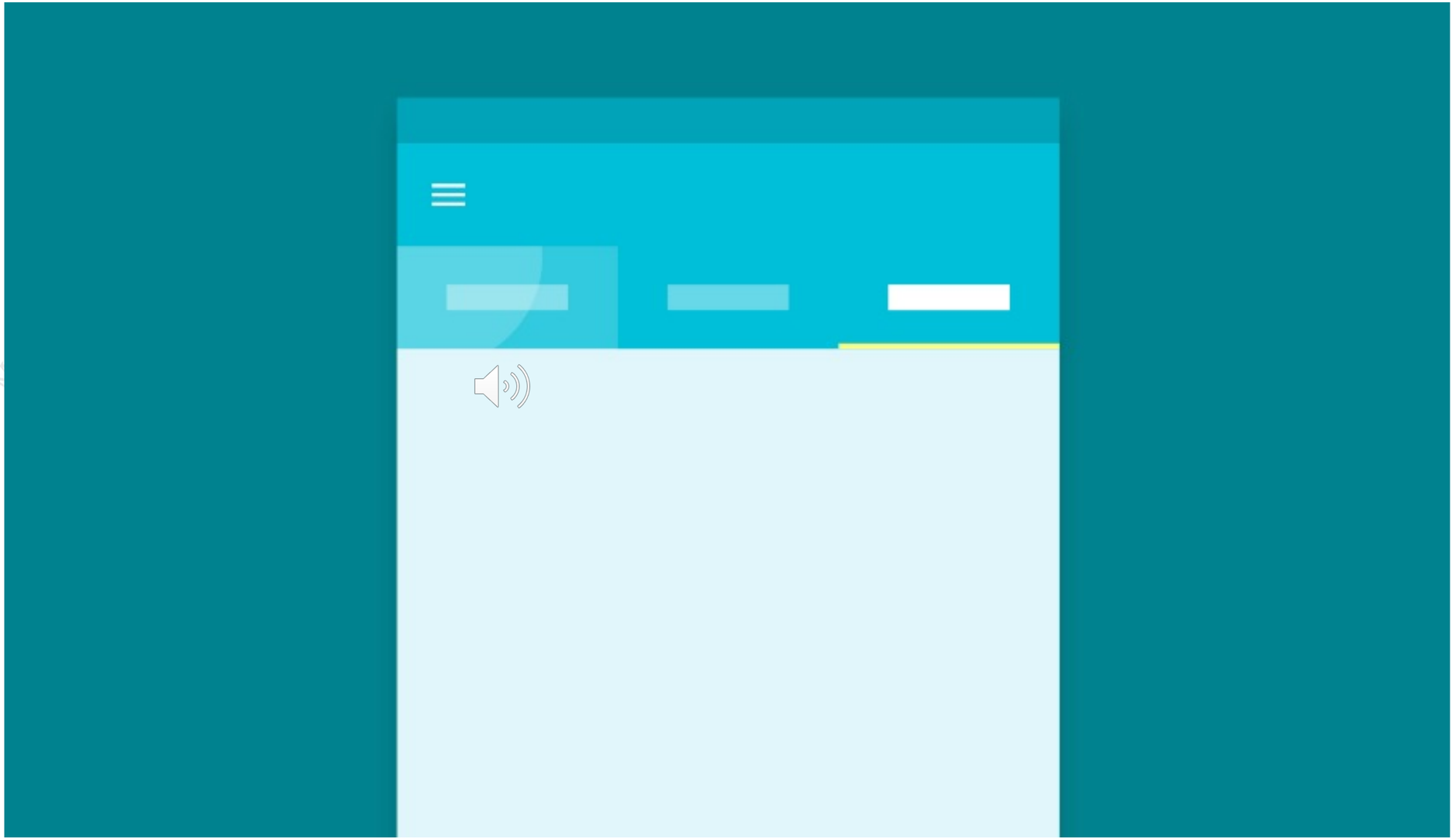
- Создавайте навигацию исходя из задач и контента;
- Создавайте навигацию для людей





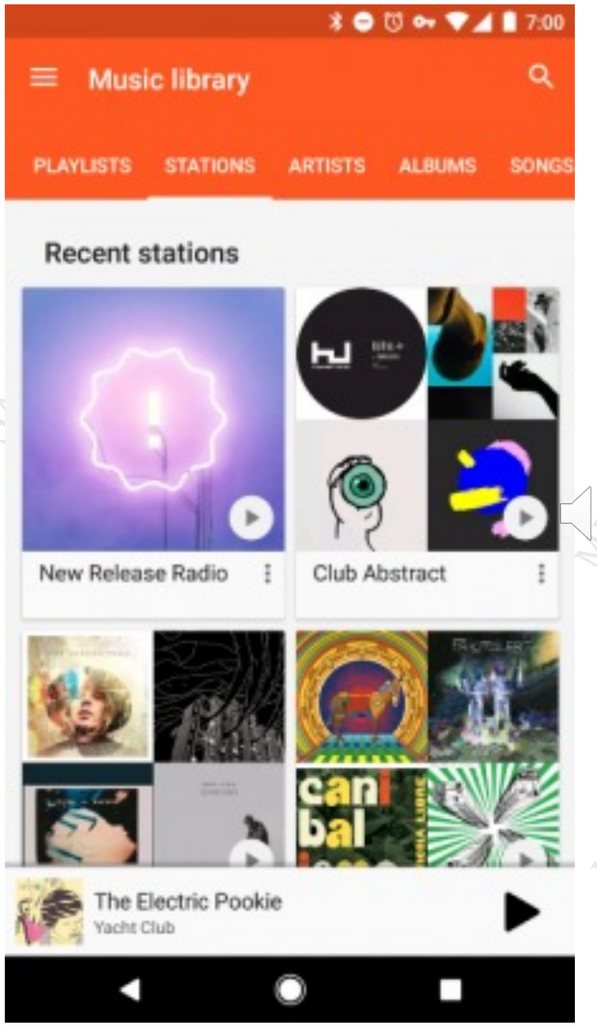
МИРЭА

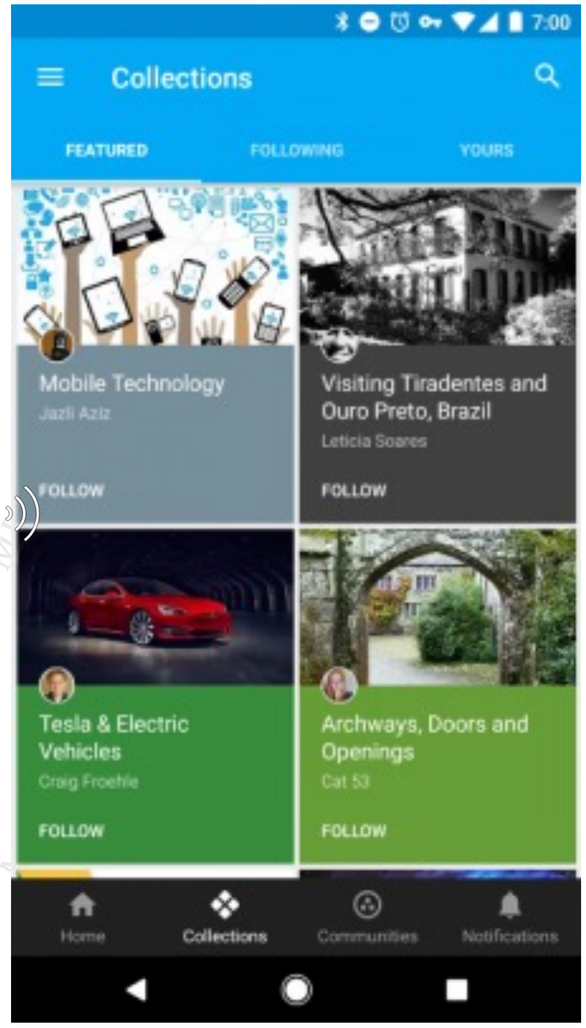
Вкладки

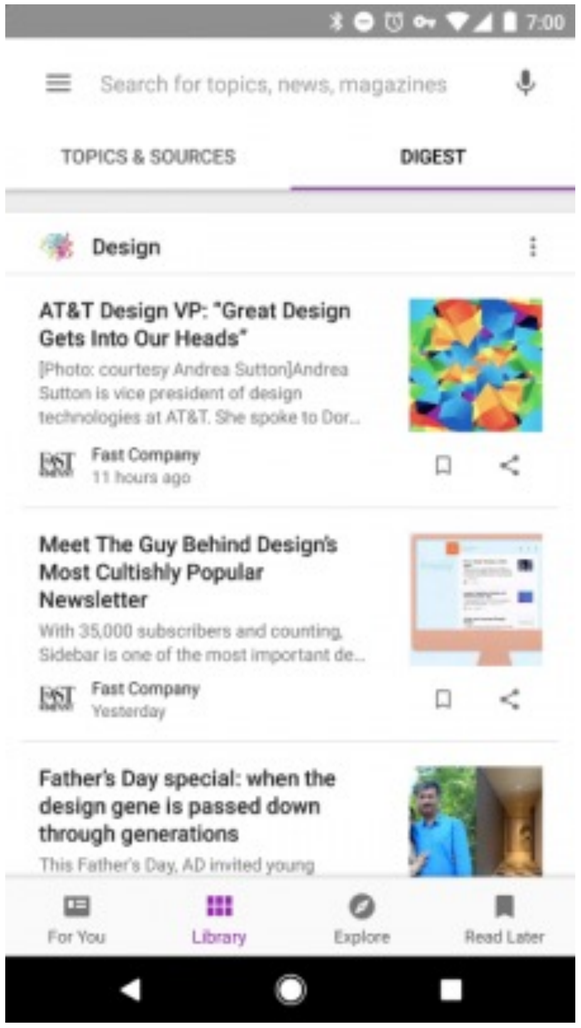


МИРЭА

Вкладки







МИРЭА

ListsLesson – DataFragment.java [ListsLesson.resultapifragment]

ListsLesson › resultapifragment › src › main › java › com › mirea › resultapifragment › DataFragment › onCreateView

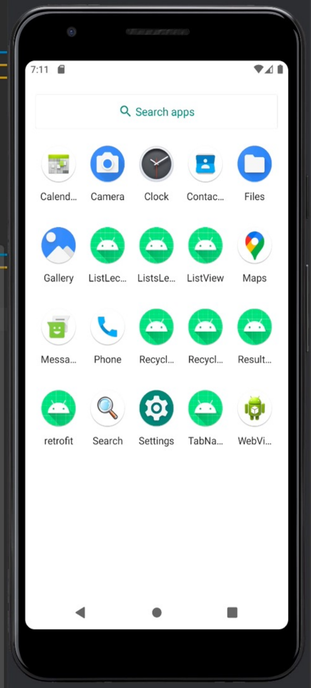
Android

- app
- listview
- recyclerview
- resultapifragment
- retrofit
- staticfragment
- Gradle Scripts

DataFragment.java

```
41      */
42     // TODO: Rename and change types and number of parameters
43     public static DataFragment newInstance(String param1, String param2) {
44         DataFragment fragment = new DataFragment();
45         Bundle args = new Bundle();
46         args.putString(ARG_PARAM1, param1);
47         args.putString(ARG_PARAM2, param2);
48         fragment.setArguments(args);
49         return fragment;
50     }
51
52     @Override
53     public void onCreate(Bundle savedInstanceState) {
54         super.onCreate(savedInstanceState);
55         if (getArguments() != null) {
56             mParam1 = getArguments().getString(ARG_PARAM1);
57             mParam2 = getArguments().getString(ARG_PARAM2);
58         }
59     }
60
61     @Override
62     public View onCreateView(LayoutInflater inflater, ViewGroup container,
63                              Bundle savedInstanceState) {
64         // Inflate the layout for this fragment
65         View view = inflater.inflate(R.layout.fragment_data, container,  attachToRoot: false);
66         Button button = view.findViewById(R.id.buttonSendData);
67
68         button.setOnClickListener(click->{
69             String text = ((TextView)view.findViewById(R.id.editTextData)).getText().toString();
70             Log.d(DataFragment.class.getSimpleName(), text);
71             Bundle bundle = new Bundle();
72             bundle.putString("key", text);
73             getChildFragmentManager().setFragmentResult( requestKey: "requestKey", bundle);
74
75             BottomSheetFragment bottomSheetFragment = BottomSheetFragment.newInstance();
76             bottomSheetFragment.show(getChildFragmentManager(),  tag: "ModalBottomSheet");
77         });
78
79
80         return view;
81     }
82 }
```

Emulator: Pixel 3a API 30 x86



Logcat

Emulator Pixel_3a_API_30_x86 | com.mirea.resultapifragment (279 | Verbose | Regex | Show only selected application

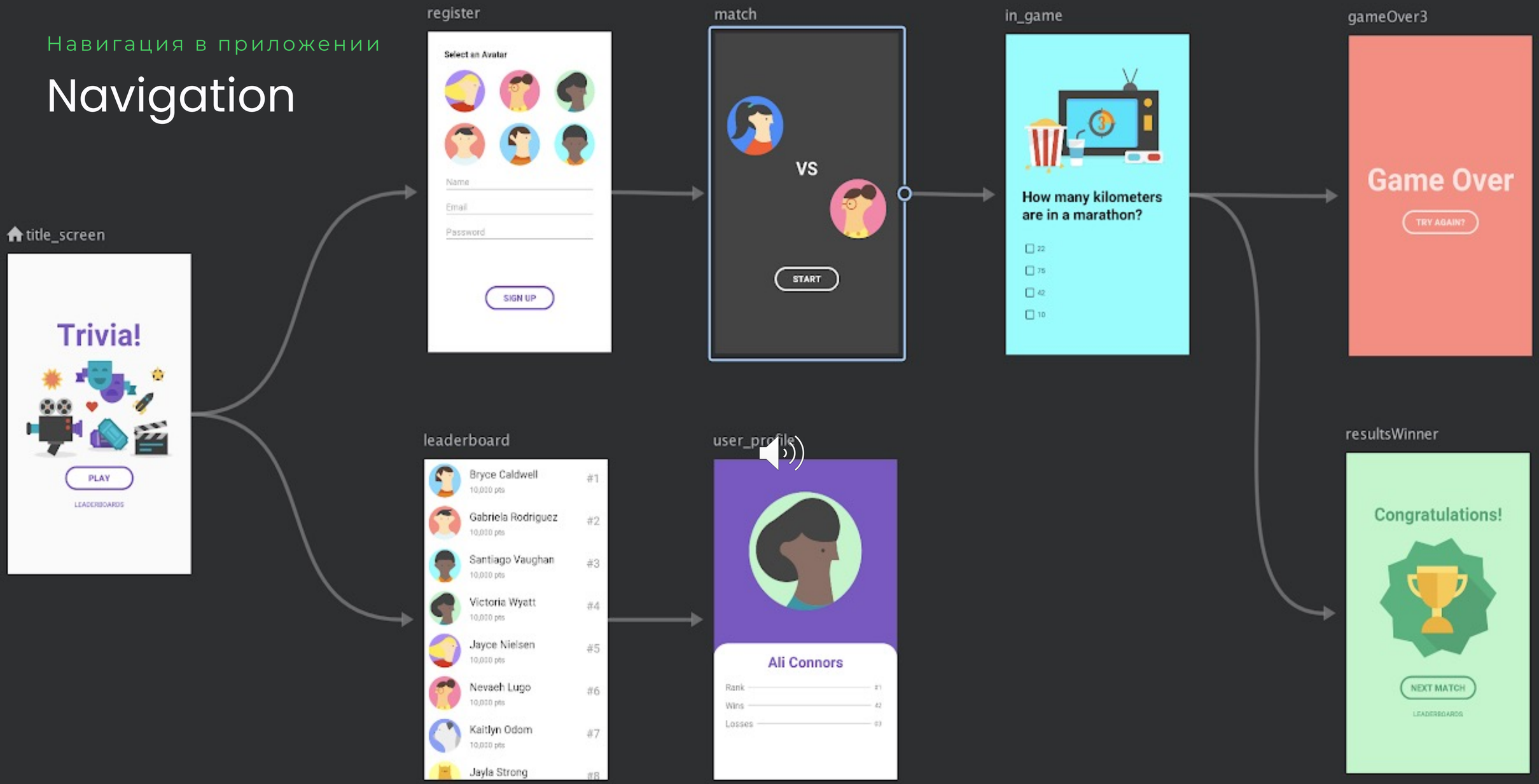
Run | TODO | Problems | Terminal | Logcat | Build | Profiler | App Inspection | Event Log | Layout Inspector

An unexpected packet was received before the handshake. (18 minutes ago)

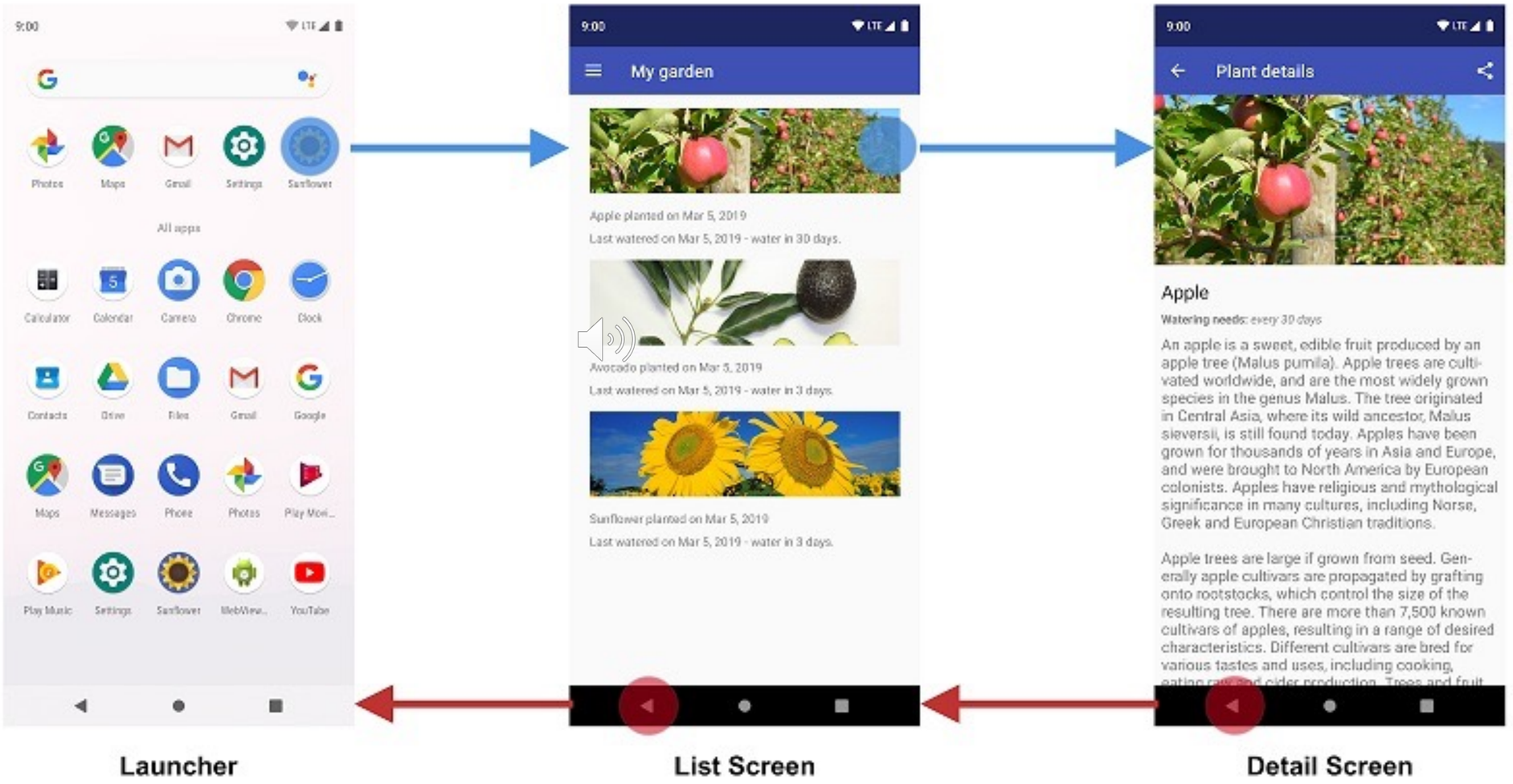
62:75 LF UTF-8 4 spaces

Навигация в приложении

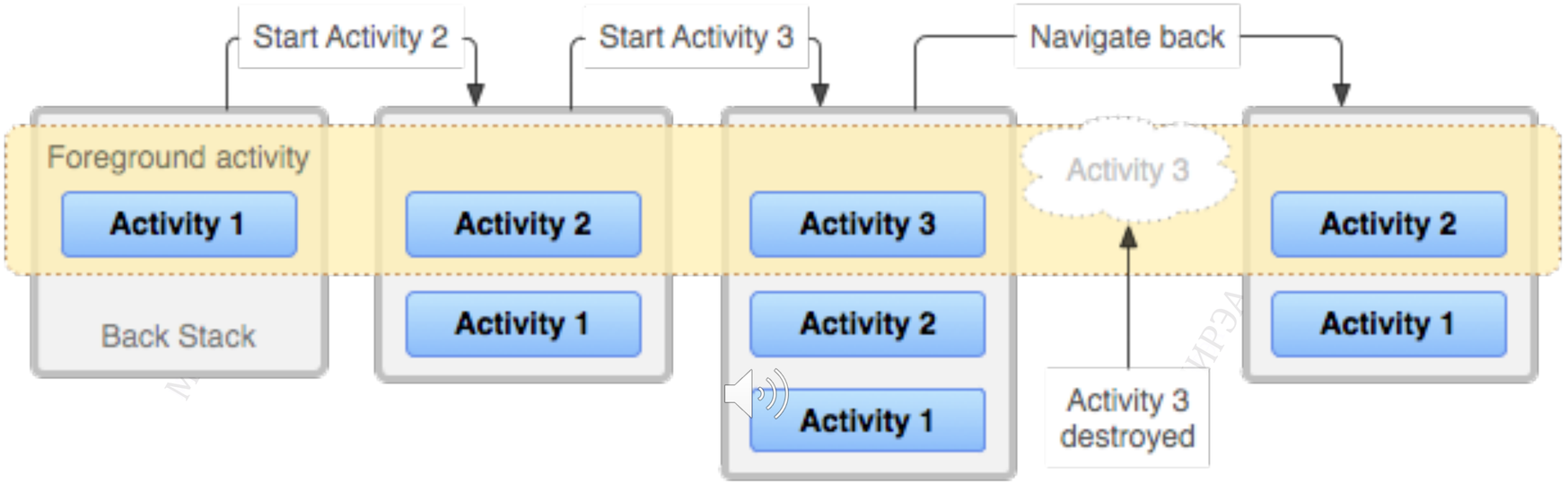
Navigation



Navigation


9:00
Photos
Maps
Gmail
Settings
Sunflower
All apps
Calculator
Calendar
Camera
Chrome
Clock
Contacts
Drive
Files
Gmail
Google
Maps
Messages
Phone
Photos
Play Movi...
Play Music
Settings
Sunflower
WebView...
YouTube
9:00
My garden
Apple planted on Mar 5, 2019
Last watered on Mar 5, 2019 - water in 30 days.
Avocado planted on Mar 5, 2019
Last watered on Mar 5, 2019 - water in 3 days.
Sunflower planted on Mar 5, 2019
Last watered on Mar 5, 2019 - water in 3 days.
9:00
Plant details
Apple
Watering needs: every 30 days
An apple is a sweet, edible fruit produced by an apple tree (Malus pumila). Apple trees are cultivated worldwide, and are the most widely grown species in the genus Malus. The tree originated in Central Asia, where its wild ancestor, Malus sieversii, is still found today. Apples have been grown for thousands of years in Asia and Europe, and were brought to North America by European colonists. Apples have religious and mythological significance in many cultures, including Norse, Greek and European Christian traditions.
Apple trees are large if grown from seed. Generally apple cultivars are propagated by grafting onto rootstocks, which control the size of the resulting tree. There are more than 7,500 known cultivars of apples, resulting in a range of desired characteristics. Different cultivars are bred for various tastes and uses, including cooking,
Launcher
List Screen
Detail Screen

Navigation





Кнопки «Вверх» и «Назад»

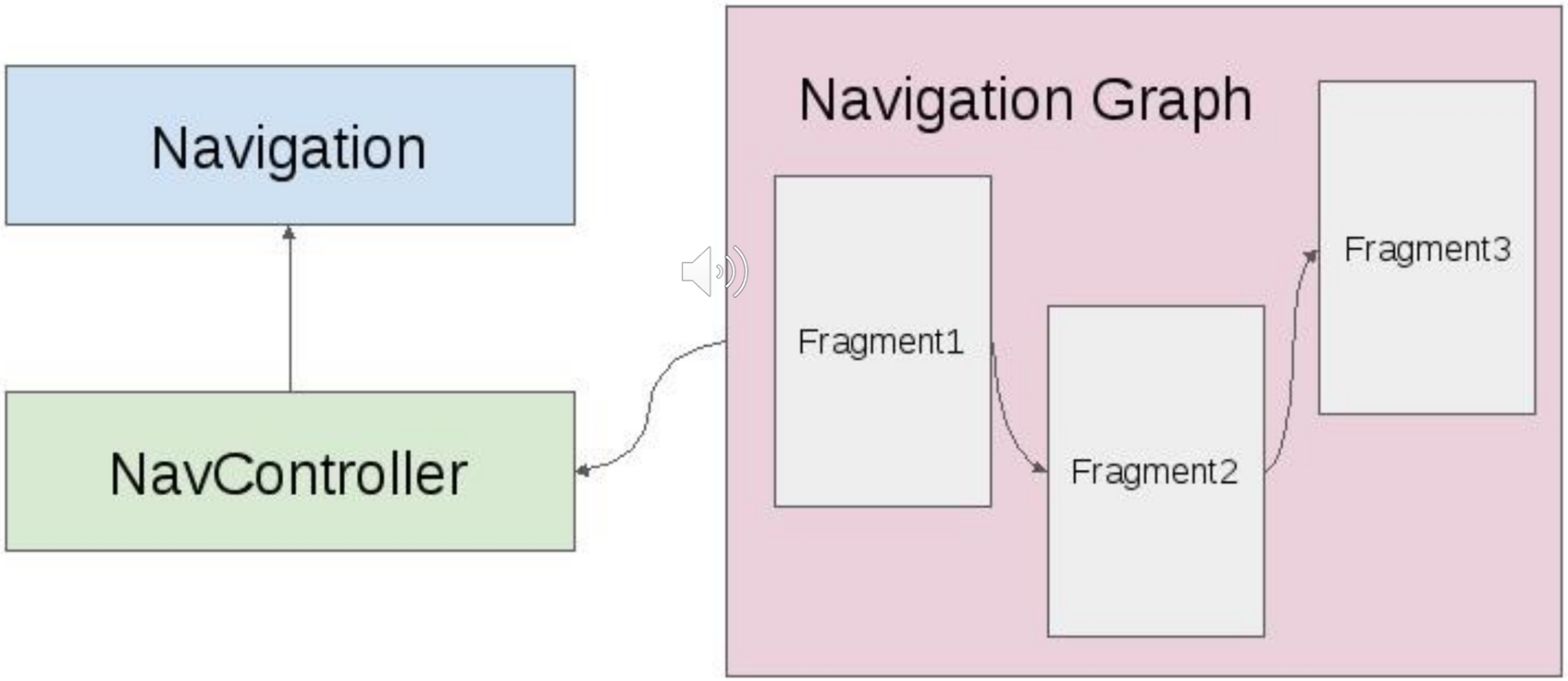
МИРЭА

Navigation Architecture Component

1. Автоматическая обработка транзакций фрагментов;
2. Корректная обработка кнопок «В**верх»** и «**Назад»** по умолчанию;
3. Поведение по умолчанию для анимации и переходов;
4. Реализация шаблонов навигации пользовательского интерфейса (таких как navigation drawer и bottom navigation) с небольшой дополнительной работой;
5. Безопасность типов при передаче информации во время навигации;
6. Инструменты Android Studio для визуализации и редактирования navigation flow приложения.

МИРЭА

Navigation Architecture Component. Структура

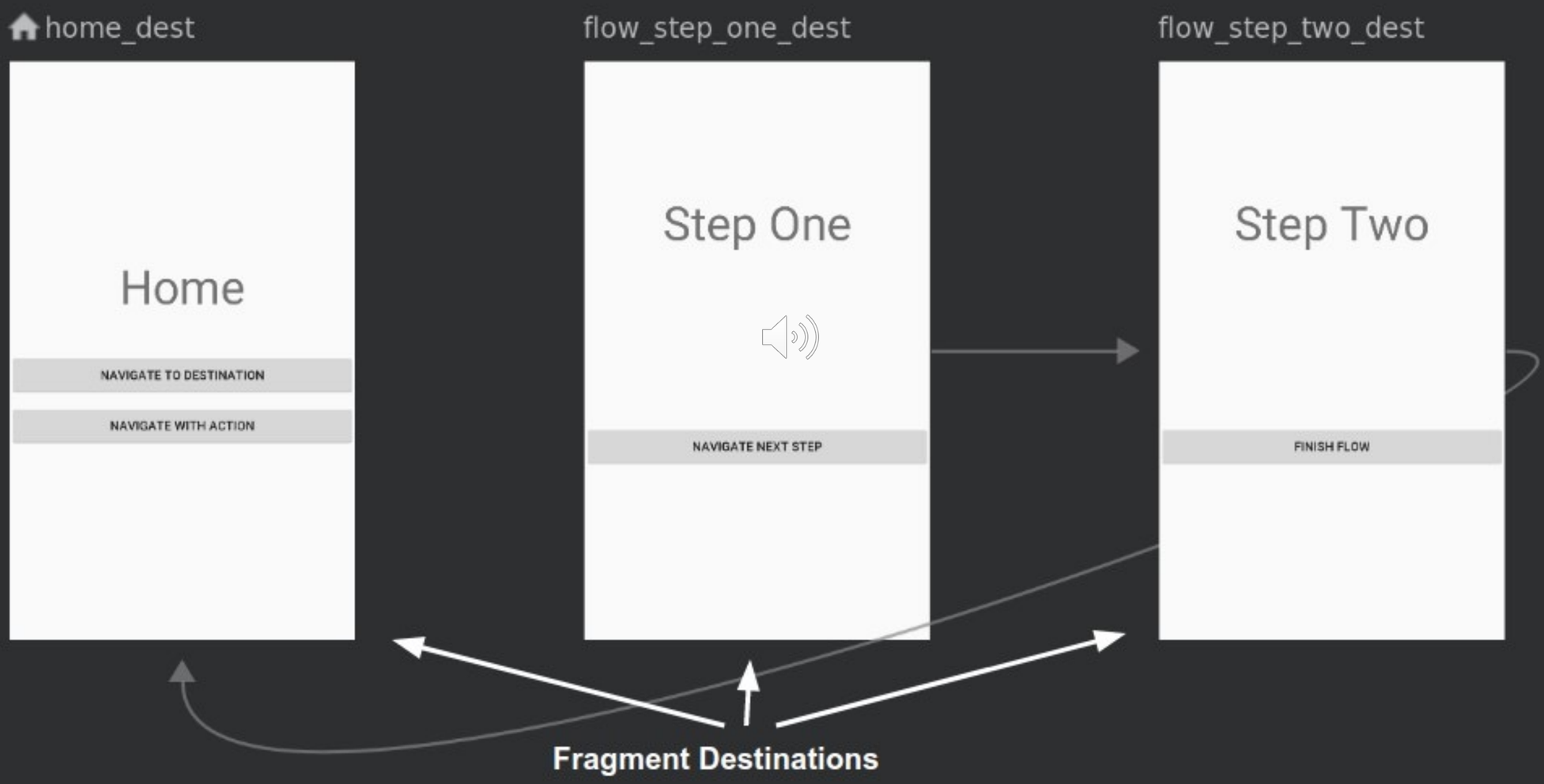


Navigation Architecture Component.

Destinations (пункт назначения) — это любое место, в котором вы можете перемещаться в приложении, обычно это фрагмент или активити. Они поддерживаются «из коробки», но вы также можете создавать свои собственные типы назначения, если это необходимо.

Navigation Graph — представляет собой новый тип ресурса, который определяет все возможные пути, доступные пользователю в приложении. Он показывает визуально все пункты назначения, которые могут быть достигнуты из данного пункта назначения. Редактор навигации Android Studio отображает Navigation Graph наглядно.

Navigation Architecture Component



Navigation Architecture Component

```
<navigation xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/mobile_navigation"
    app:startDestination="@+id/home_dest">

    <fragment
        android:id="@+id/home_dest"
        android:name="com.mirea.navigationarchcomponent.HomeFragment"
        android:label="Home"
        tools:layout="@layout/fragment_home">
        <action
            android:id="@+id/next_action"
            app:destination="@+id/flowStepOne" />
```

МИРЭА

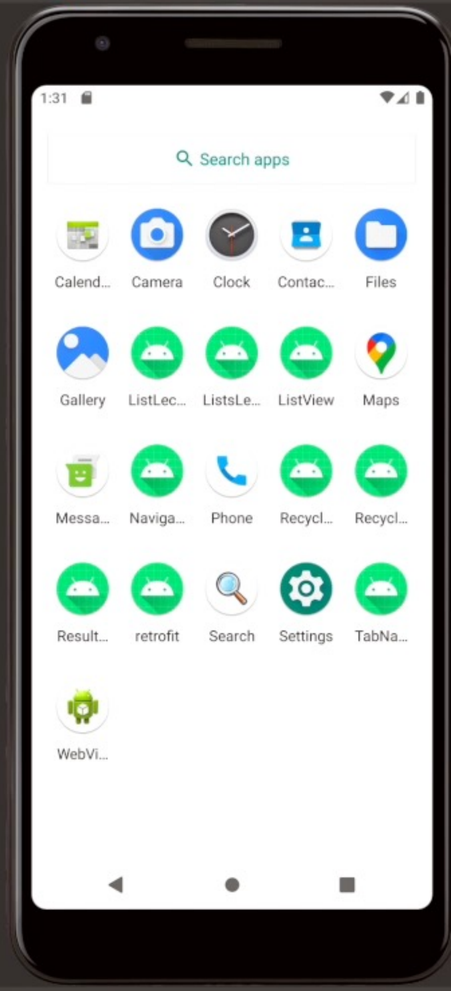
```java
package com.mirea.tabnavigation;

import ...

public class MainActivity extends FragmentActivity {
    private NumberAdapter adapter;
    private ViewPager2 viewPager;
    String[] tabNames = new String[]{"Первый", "Второй", "Третий", "Четвертый", "Пятый", "Шестой"};
    int[] tabImages = new int[] {R.drawable.ic_baseline_looks_one_24,
            R.drawable.ic_baseline_looks_two_24,
            R.drawable.ic_baseline_looks_3_24,
            R.drawable.ic_baseline_looks_4_24,
            R.drawable.ic_baseline_looks_5_24,
            R.drawable.ic_baseline_looks_6_24 };
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

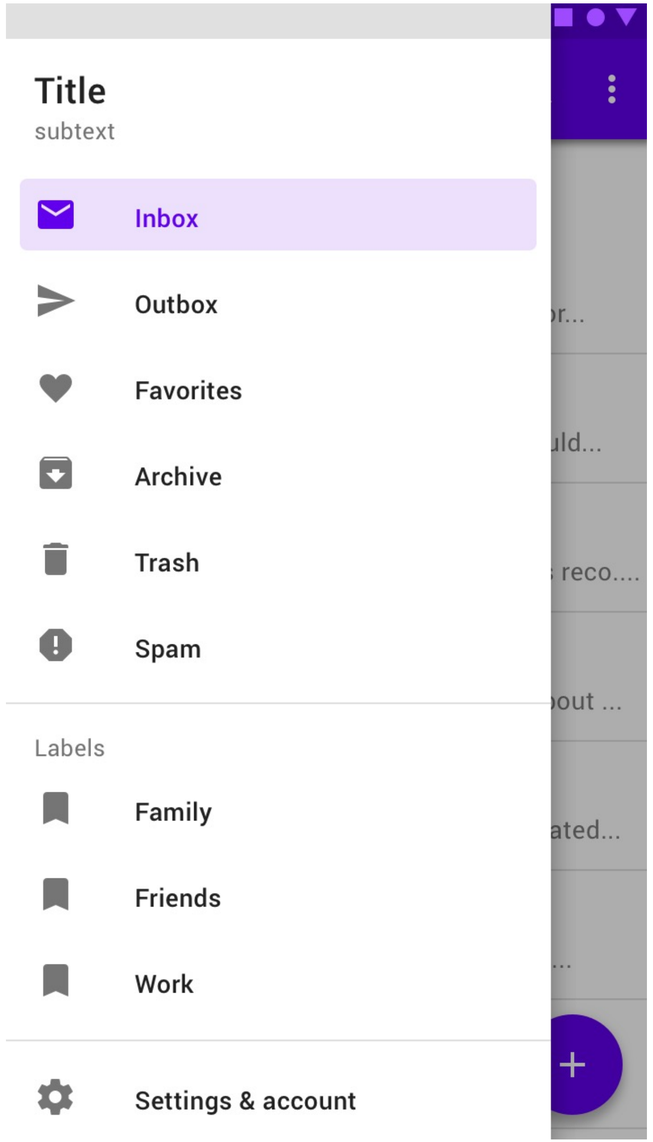
        adapter = new NumberAdapter( fragmentActivity: this);
        viewPager = findViewById(R.id.viewPager);
        viewPager.setAdapter(adapter);

        TabLayout tabLayout = findViewById(R.id.tabLayout);
        new TabLayoutMediator(tabLayout, viewPager, new TabLayoutMediator.TabConfigurationStrategy() {
            @Override
            public void onConfigureTab(@NonNull TabLayout.Tab tab, int position) {
                tab.setText(tabNames[position]);
                tab.setIcon(tabImages[position]);
                if(position == 1){
                    BadgeDrawable badgeDrawable = tab.getOrCreateBadge();
                    badgeDrawable.setNumber(2);
                }
            }
        }).attach();
    }
}
```

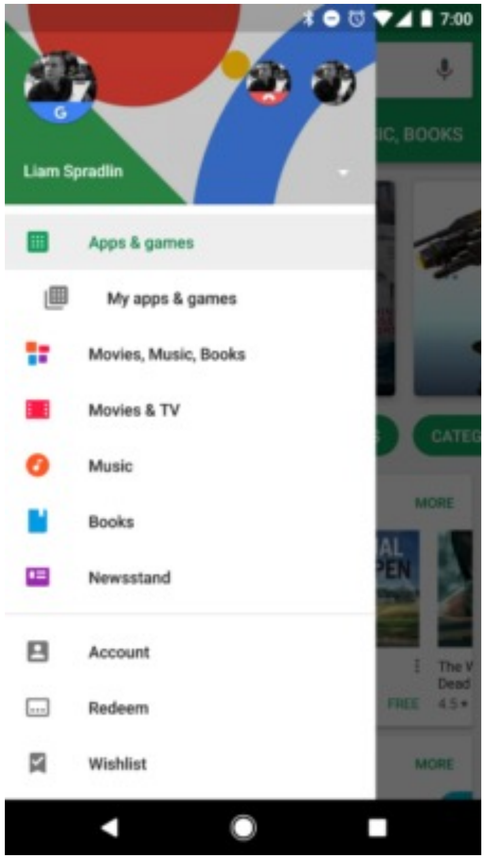

```
2022-02-09 01:01:48.719 0-0/? I/init: processing action (sys.init.updatable_crashing=1) from (/system/etc/init/flags_health_check.rc:10)
2022-02-09 01:01:48.721 0-0/? I/init: starting service 'exec 6037 (/system/bin/flags_health_check UPDATABLE_CRASHING)'...
2022-02-09 01:01:48.723 0-0/? I/init: SVC_EXEC service 'exec 6037 (/system/bin/flags_health_check UPDATABLE_CRASHING)' pid 18447 (uid 1000 gid 1000+0 context default) started; waiting...
2022-02-09 01:01:48.738 18447-18447/? I/flags_health_check: ServerConfigurableFlagsReset reset_mode value: 1
2022-02-09 01:01:48.738 18447-18447/? I/flags_health_check: ServerConfigurableFlagsReset updatable crashing detected, resetting flags.
2022-02-09 01:01:48.728 0-0/? I/init: Service 'exec 6037 (/system/bin/flags_health_check UPDATABLE_CRASHING)' (pid 18447) exited with status 0 waiting took 0.005000 seconds
2022-02-09 01:01:48.730 0-0/? I/init: Sending signal 9 to service 'exec 6037 (/system/bin/flags_health_check UPDATABLE_CRASHING)' (pid 18447) process group...
2022-02-09 01:01:48.732 0-0/? I/libprocessgroup: Successfully killed process cgroup uid 1000 pid 18447 in 0ms
2022-02-09 01:01:53.710 0-0/? I/init: starting service 'console'...
2022-02-09 01:01:53.717 0-0/? I/init: Service 'console' (pid 18668) exited with status 0
```

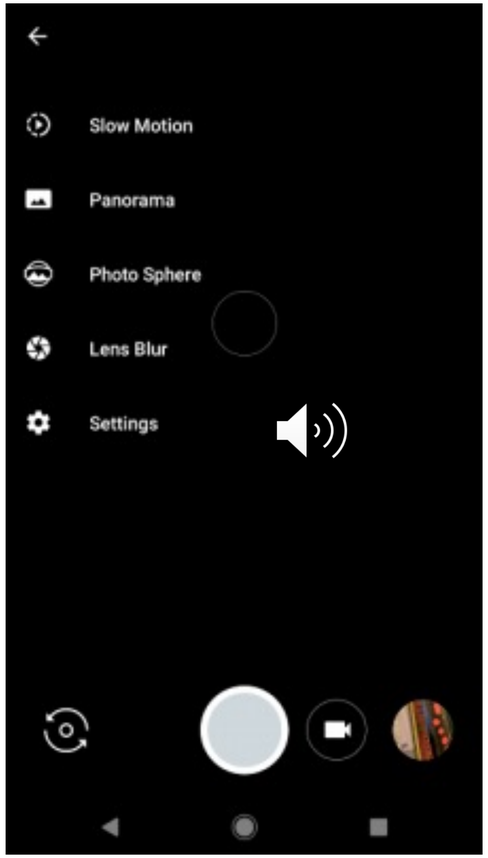
Navigation Drawer

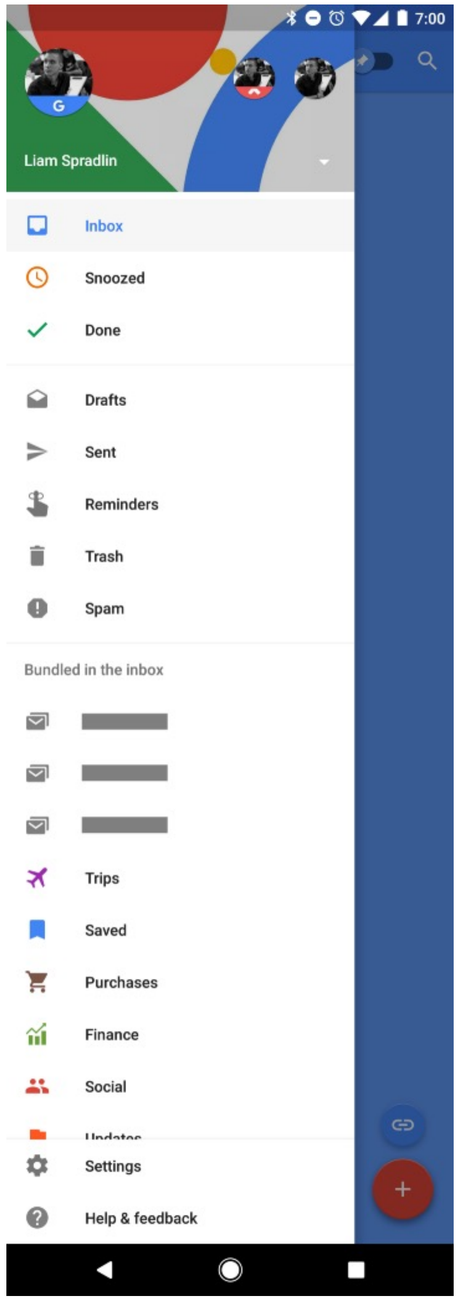


МИРЭА

Navigation Drawer







МИРЭА

ListsLesson – mobile_navigation.xml [ListsLesson.navigationarchcomponent]

son › navigationarchcomponent › src › main › res › navigation › mobile_navigation.xml

navigationarchcomponent | Pixel_3a_API_30_x86

MainActivity.java | mobile_navigation.xml

Code | Split | Design

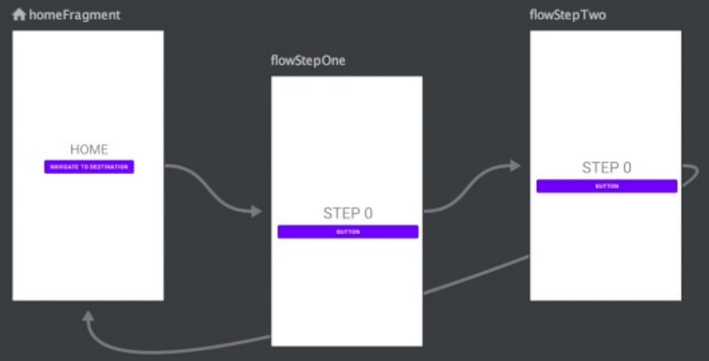
Emulator: Pixel 3a API 30 x86

droid

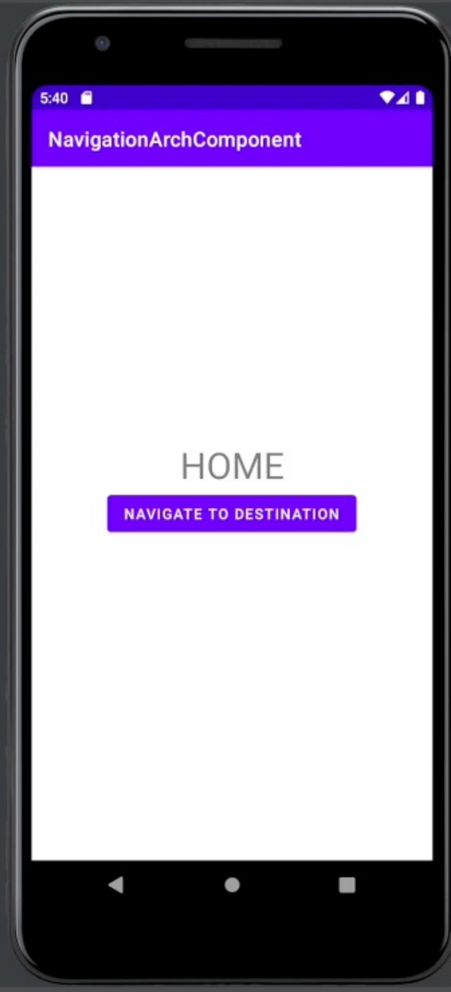
app
listview
navigationarchcomponent
manifests
java
com.mirea.navigationarchcomponent
FlowStepFragment
HomeFragment
MainActivity
com.mirea.navigationarchcomponent (a
com.mirea.navigationarchcomponent (t
java (generated)
res
drawable
layout
activity_main.xml
fragment_flow_step.xml
fragment_home.xml
mipmap
navigation
mobile_navigation.xml
values
res (generated)
recyclerview
resultapifragment
retrofit
staticfragment
tabnavigation
Gradle Scripts
build.gradle (Project: ListsLesson)
build.gradle (Module: ListsLesson.app)
build.gradle (Module: ListsLesson.listview)
build.gradle (Module: ListsLesson.navigati
build.gradle (Module: ListsLesson.recycler
build.gradle (Module: ListsLesson.resultap
build.gradle (Module: ListsLesson.retrofit)
build.gradle (Module: ListsLesson.staticfra
build.gradle (Module: ListsLesson.tabnavig
gradle-wrapper.properties (Gradle Version
proguard-rules.pro (ProGuard Rules for Lis
proguard-rules.pro (ProGuard Rules for Lis
proguard-rules.pro (ProGuard Rules for Lis
proguard-rules.pro (ProGuard Rules for Lis
proguard-rules.pro (ProGuard Rules for Lis
proguard-rules.pro (ProGuard Rules for Lis
proguard-rules.pro (ProGuard Rules for Lis
proguard-rules.pro (ProGuard Rules for Lis
gradle.properties (Project Properties)

Hosts
activity_main (nav_host_fragme

Component Tree
mobile_navigation - navigati
homeFragment - fragmen
flowStepOne - fragment
flowStepTwo - fragment



Attributes



mulator Pixel_3a_API_30_x86 | No debuggable processes | Verbose | Regex | Show only selected application

```
2022-02-09 17:40:18.525 0-0/? E/init: updatable process 'console' exited 4 times in 4 minutes
2022-02-09 17:40:18.528 0-0/? I/init: processing action (sys.init.updatable_crashing=1) from (/system/etc/init/flags_health_check.rc:10)
2022-02-09 17:40:18.530 0-0/? I/init: starting service 'exec 3032 (/system/bin/flags_health_check UPDATABLE_CRASHING)'...
2022-02-09 17:40:18.531 0-0/? I/init: SVC_EXEC service 'exec 3032 (/system/bin/flags_health_check UPDATABLE_CRASHING)' pid 15982 (uid 1000 gid 1000+0 context default) started; waiting...
2022-02-09 17:40:18.557 15982-15982/? I/flags_health_check: ServerConfigurableFlagsReset reset_mode value: 1
2022-02-09 17:40:18.558 15982-15982/? I/flags_health_check: ServerConfigurableFlagsReset updatable crashing detected, resetting flags.
2022-02-09 17:40:18.539 0-0/? I/init: Service 'exec 3032 (/system/bin/flags_health_check UPDATABLE_CRASHING)' (pid 15982) exited with status 0 waiting took 0.007000 seconds
2022-02-09 17:40:18.541 0-0/? I/init: Sending signal 9 to service 'exec 3032 (/system/bin/flags_health_check UPDATABLE_CRASHING)' (pid 15982) process group...
2022-02-09 17:40:18.543 0-0/? I/libprocessgroup: Successfully killed process cgroup uid 1000 pid 15982 in 0ms
```

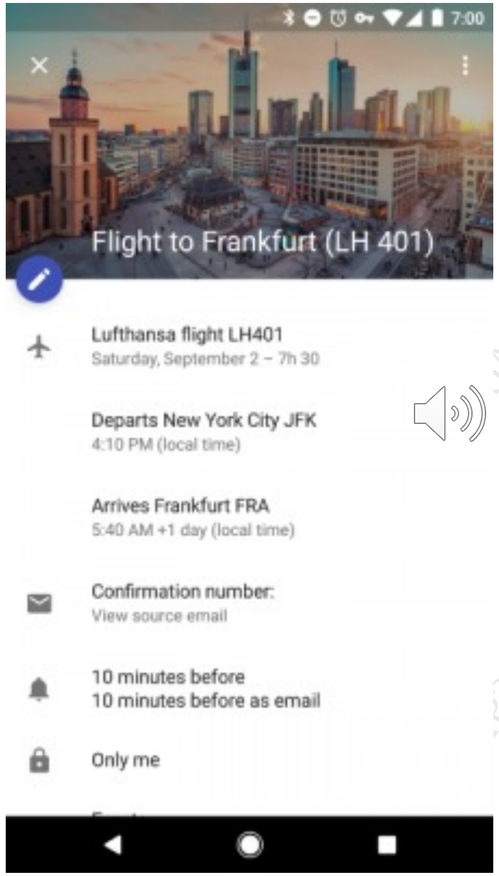
TODO | Problems | Terminal | Logcat | Build | Profiler | App Inspection | Event Log | Layout Ins

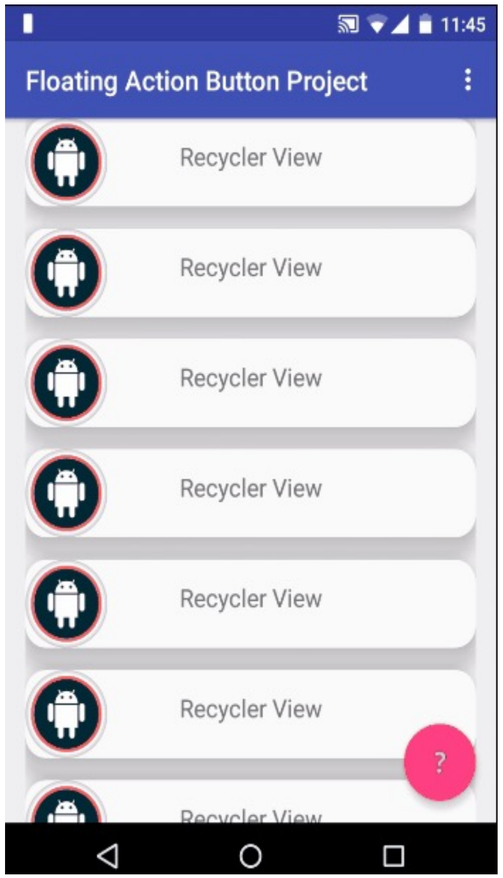
succeded (5 minutes ago)

3:28 LF UTF-8 4 spaces

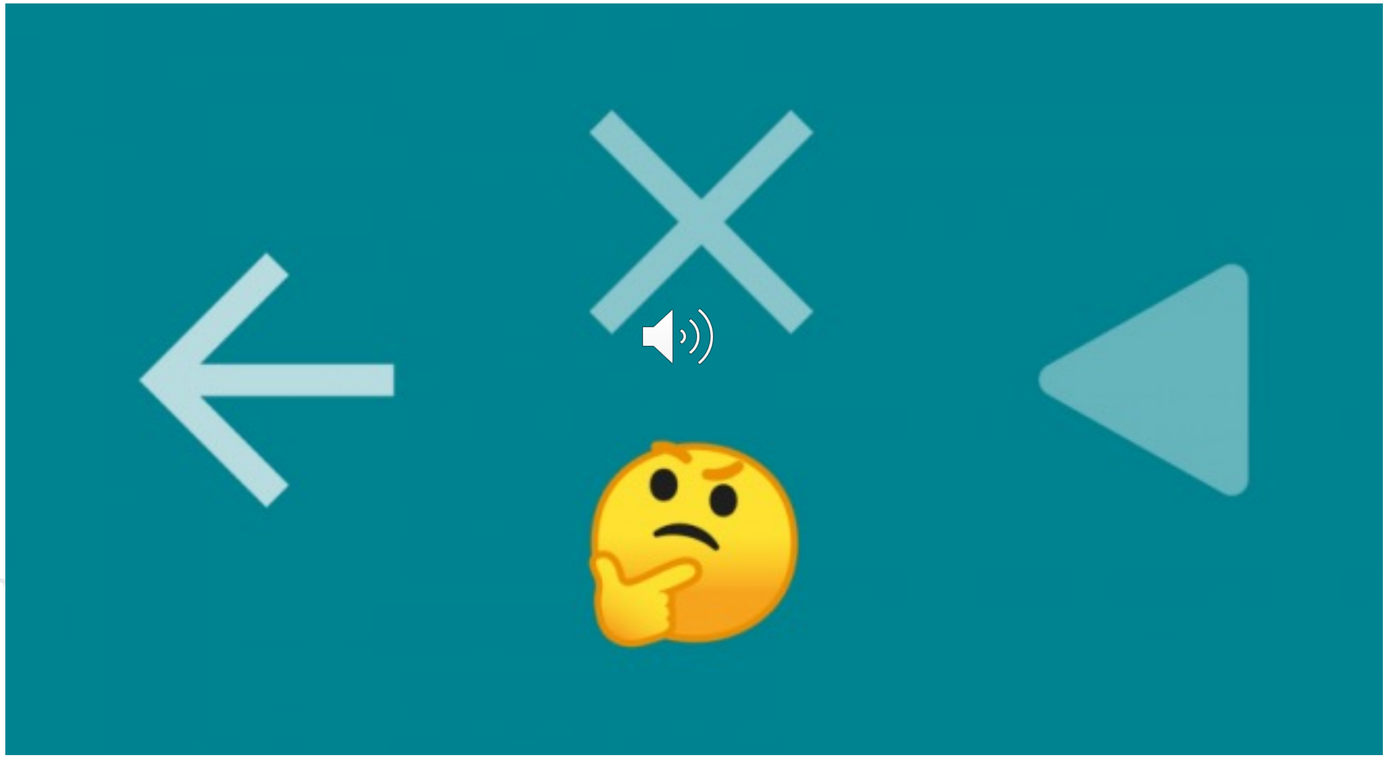
Контекстная навигация







Контекстная навигация



МИРЭА

МИРЭА

ListsLesson – activity_main.xml [ListsLesson.navigationarchcomponent]

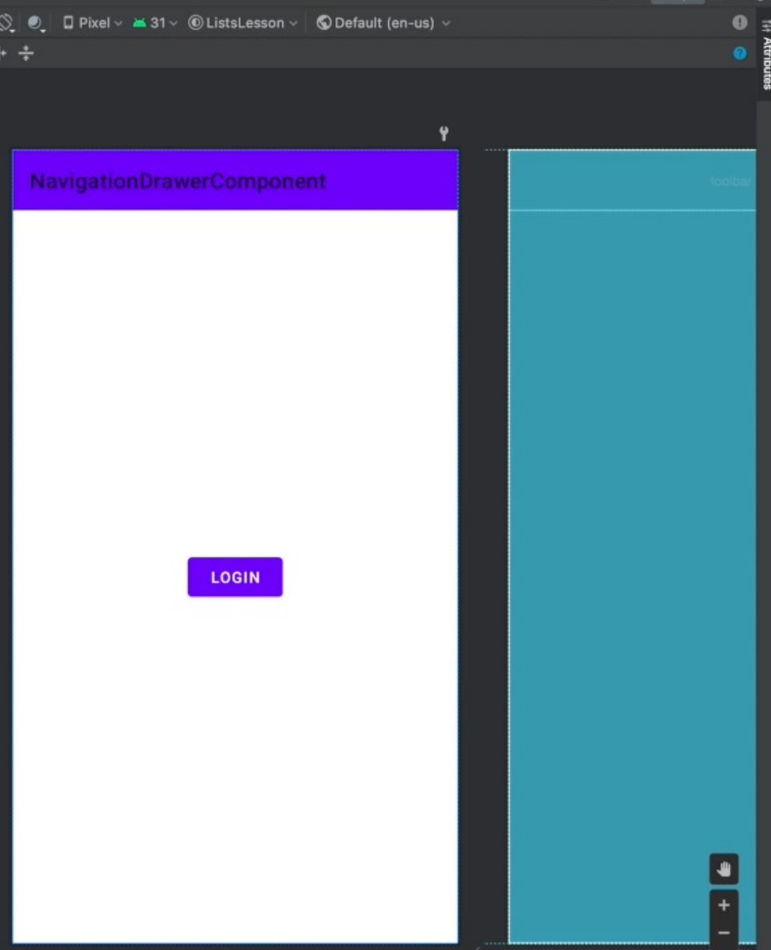
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.drawerlayout.widget.DrawerLayout xmlns:android="http://schemas.
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:id="@+id/drawer"
    tools:context=".MainActivity">

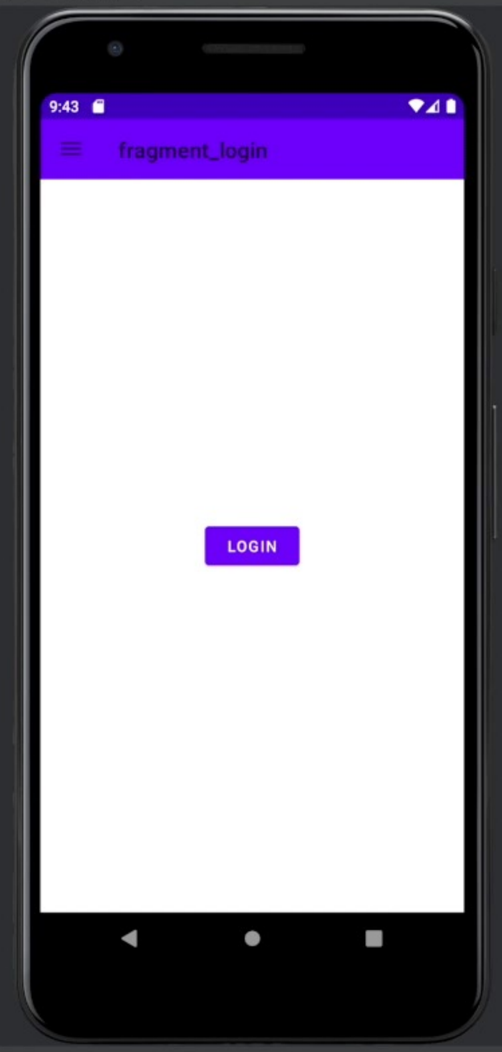

    <androidx.appcompat.widget.LinearLayoutCompat
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical">

        <androidx.appcompat.widget.Toolbar
            android:id="@+id/toolbar"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:minHeight="?attr/actionBarSize"
            android:background="@color/purple_500"
            app:title="NavigationDrawerComponent"/>

        <androidx.fragment.app.FragmentContainerView
            android:id="@+id/nav_host_fragment"
            android:name="androidx.navigation.fragment.NavHostFragment"
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            app:defaultNavHost="true"
            app:layout_constraintBottom_toBottomOf="parent"
            app:layout_constraintLeft_toLeftOf="parent"
            app:layout_constraintRight_toRightOf="parent"
            app:layout_constraintTop_toTopOf="parent"
            app:navGraph="@navigation/mobile_navigation" />

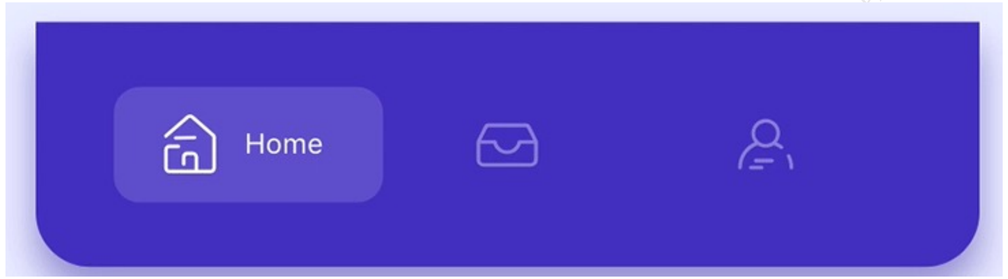
    </androidx.appcompat.widget.LinearLayoutCompat>
    <com.google.android.material.navigation.NavigationView
        android:id="@+id/nav_view"
        android:fitsSystemWindows="true"
        android:layout_gravity="start"
        app:menu="@menu/menu"
        app:headerLayout="@layout/nav_header"
        android:layout_width="wrap_content"
        android:layout_height="match_parent" />
```






```
2022-02-09 21:43:12.750 0-0/? I/init: processing action (sys.init.updatable_crashing=1) from (/system/etc/init/flags_health_check.rc:10)
2022-02-09 21:43:12.752 0-0/? I/init: starting service 'exec 5699 (/system/bin/flags_health_check UPDATABLE_CRASHING)'...
2022-02-09 21:43:12.776 28413-28413/? I/flags_health_check: ServerConfigurableFlagsReset reset_mode value: 1
2022-02-09 21:43:12.776 28413-28413/? I/flags_health_check: ServerConfigurableFlagsReset updatable crashing detected, resetting flags.
2022-02-09 21:43:12.754 0-0/? I/init: SVC_EXEC service 'exec 5699 (/system/bin/flags_health_check UPDATABLE_CRASHING)' pid 28413 (uid 1000 gid 1000+0 context default) started; waiting...
2022-02-09 21:43:12.759 0-0/? I/init: Service 'exec 5699 (/system/bin/flags_health_check UPDATABLE_CRASHING)' (pid 28413) exited with status 0 waiting took 0.005000 seconds
2022-02-09 21:43:12.761 0-0/? I/init: Sending signal 9 to service 'exec 5699 (/system/bin/flags_health_check UPDATABLE_CRASHING)' (pid 28413) process group...
2022-02-09 21:43:12.763 0-0/? I/libprocessgroup: Successfully killed process cgroup uid 1000 pid 28413 in 0ms
```

BottomNavigationView



МИРЭА

ListsLesson › navigationarchcomponent › src › main › java › com › mirea › navigationarchcomponent › MainActivity › onCreate

navigationarchcomponent | Pixel_3a_API_30_x86

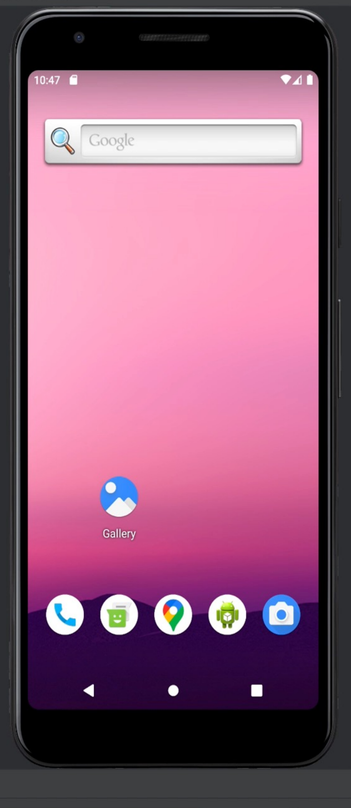
Android

- app
- listview
- navigationarchcomponent
  - manifests
  - java
    - com.mirea.navigationarchcomponent
    - com.mirea.navigationarchcomponent (a
    - com.mirea.navigationarchcomponent (t
  - java (generated)
  - res
    - drawable
    - layout
      - activity_main.xml
      - app_bar_main.xml
      - content_main.xml
      - fragment_flow_step.xml
      - fragment_home.xml
      - fragment_login.xml
      - nav_header.xml
    - menu
    - mipmap
    - navigation
      - mobile_navigation.xml
    - values
    - xml
  - res (generated)
- recyclerview
- resultapifragment
- retrofit
- staticfragment
- tabnavigation
- Gradle Scripts
  - build.gradle (Project: ListsLesson)
  - build.gradle (Module: ListsLesson.app)
  - build.gradle (Module: ListsLesson.listview)
  - build.gradle (Module: ListsLesson.navigati
  - build.gradle (Module: ListsLesson.recycler

MainActivity.java | menu.xml | night/themes.xml | activity_main.xml | content_main.xml | app_bar_main.xml | nav_header.xml | HomeFragment.java

```
28      private AppBarConfiguration appBarConfiguration;
29      @Override
30      protected void onCreate(Bundle savedInstanceState) {
31          super.onCreate(savedInstanceState);
32          setContentView(R.layout.activity_main);
33          drawerLayout = findViewById(R.id.drawer);
34
35          Toolbar toolbar = findViewById(R.id.toolbar);
36          setSupportActionBar(toolbar);
37
38          FloatingActionButton fab = findViewById(R.id.floatingActionButton);
39          fab.setOnClickListener(view -> {
40              Snackbar.make(view, text: "Pushed fab", Snackbar.LENGTH_LONG).setAction( text: "Action", listener: null).show();
41              navController.navigate(R.id.settingsFragment, args: null);
42          });
43
44          navHostFragment = (NavHostFragment) getSupportFragmentManager().findFragmentById(R.id.nav_host_fragment);
45          navController = navHostFragment != null ? navHostFragment.getNavController() : null;
46
47          NavigationView navigationView = findViewById(R.id.nav_view);
48
49          appBarConfiguration = new AppBarConfiguration.Builder(R.id.navigation_login, R.id.navigation_home).setOpenableLayout(drawerLayout).build();
50          NavigationUI.setupActionBarWithNavController( activity: this, navController, appBarConfiguration);
51          NavigationUI.setupWithNavController(navigationView, navController);
52
53      }
54
55      @Override
56      public void onBackPressed() {
57          if (drawerLayout.isDrawerOpen(GravityCompat.START)) {
58              drawerLayout.closeDrawer(GravityCompat.START);
59          } else if (getSupportFragmentManager().getBackStackEntryCount() > 0) {
60              getSupportFragmentManager().popBackStack();
61          } else {
```

Emulator: Pixel 3a API 30 x86



Logcat

Emulator Pixel_3a_API_30_x86 | No debuggable processes | Verbose | Regex | Show only selected application

```
2022-02-09 22:10:49.990 0-0/? I/init: Sending signal 9 to service 'console' (pid 29547) process group...
2022-02-09 22:10:49.991 0-0/? I/libprocessgroup: Successfully killed process cgroup uid 2000 pid 29547 in 0ms
2022-02-09 22:10:49.992 0-0/? E/init: updatable process 'console' exited 4 times in 4 minutes
2022-02-09 22:10:49.994 0-0/? I/init: processing action (sys.init.updatable_crashing=1) from (/system/etc/init/flags_health_check.rc:10)
2022-02-09 22:10:49.996 0-0/? I/init: starting service 'exec 6002 (/system/bin/flags_health_check UPDATABLE_CRASHING)'...
2022-02-09 22:10:49.998 0-0/? I/init: SVC_EXEC service 'exec 6002 (/system/bin/flags_health_check UPDATABLE_CRASHING)' pid 29548 (uid 1000 gid 1000+0 context default) started; waiting...
2022-02-09 22:10:50.021 29548-29548/? I/flags_health_check: ServerConfigurableFlagsReset reset_mode value: 1
2022-02-09 22:10:50.021 29548-29548/? I/flags_health_check: ServerConfigurableFlagsReset updatable crashing detected, resetting flags.
2022-02-09 22:10:50.004 0-0/? I/init: Service 'exec 6002 (/system/bin/flags_health_check UPDATABLE_CRASHING)' (pid 29548) exited with status 0 waiting took 0.006000 seconds
2022-02-09 22:10:50.006 0-0/? I/init: Sending signal 9 to service 'exec 6002 (/system/bin/flags_health_check UPDATABLE_CRASHING)' (pid 29548) process group...
2022-02-09 22:10:50.008 0-0/? I/libprocessgroup: Successfully killed process cgroup uid 1000 pid 29548 in 0ms
2022-02-09 22:10:54.992 0-0/? I/init: starting service 'console'...
2022-02-09 22:10:55.000 0-0/? I/init: Service 'console' (pid 29549) exited with status 0
2022-02-09 22:10:55.001 0-0/? I/init: Sending signal 9 to service 'console' (pid 29549) process group...
2022-02-09 22:10:55.001 0-0/? I/libprocessgroup: Successfully killed process cgroup uid 2000 pid 29549 in 0ms
2022-02-09 22:10:55.003 0-0/? E/init: updatable process 'console' exited 4 times in 4 minutes
2022-02-09 22:10:55.005 0-0/? I/init: processing action (sys.init.updatable_crashing=1) from (/system/etc/init/flags_health_check.rc:10)
2022-02-09 22:10:55.007 0-0/? I/init: starting service 'exec 6003 (/system/bin/flags_health_check UPDATABLE_CRASHING)'...
2022-02-09 22:10:55.009 0-0/? I/init: SVC_EXEC service 'exec 6003 (/system/bin/flags_health_check UPDATABLE_CRASHING)' pid 29550 (uid 1000 gid 1000+0 context default) started; waiting...
2022-02-09 22:10:55.032 29550-29550/? I/flags_health_check: ServerConfigurableFlagsReset reset_mode value: 1
2022-02-09 22:10:55.032 29550-29550/? I/flags_health_check: ServerConfigurableFlagsReset updatable crashing detected, resetting flags.
```